Harsh Gupta | 22IM3FP29

INDUSTRIAL ENGINEERING / FINANCIAL ENGINEERING
MINOR in COMPUTER SCIENCE & ENGG.

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH Dual Degree 5Y	Indian Institute of Technology Kharagpur	8.53/10
2022	Higher Secondary Education, CBSE Board	National High School	91.2%
2020	Secondary Education, CBSE Board	National High School	91%

WORK EXPERIENCE

Full Stack Engineer Intern | Chainwide, United States **Jul 2024 – Dec 2024**
- Engineered data integration platform connecting 3 systems: Salesforce, Confluence, and Notion REST APIs, automated daily ETL processes
- Developed Slack bot with Pydantic AI, handling 100K daily queries through 3 specialised RAG agents for Sales, Service, and Marketing
- Created monitoring dashboard tracking 15+ metrics for RAG agent efficiency and real-time user engagement, improving response by 40%

Research Intern | Deakin University, Australia **Mar 2024 – Jul 2024**
- Analysed 3+ research papers on anomaly detection, synthesising methodologies to inform algorithm selection for quality control
- Implemented and validated the PatchCore algorithm for manufacturing anomaly detection by achieving 99.6% AUROC on the MVTec AD
- Authored a comprehensive 8-page technical report evaluating 4+ quality control constraints and anomaly detection in manufacturing

Machine Learning Engineer Intern | Practworks, India **Dec 2023 – Mar 2024**
- Constructed a concept dependency graph for 500+ topics, used by the ML engine to optimize sequences for personalized learning paths
- Generated 67,000 synthetic data points to train robust student performance prediction models, enhancing personalised learning accuracy
- Engineered a predictive ML recommendation system using Random Forest, achieving 67% accuracy in predicting student learning gaps

Software Engineer Intern | Ylytic, India **Sep 2023 – Nov 2023**
- Integrated 5 crucial new analytics features, including engagement metrics and brand mentions, enhancing influencer evaluation capabilities
- Reduced data processing time by 60%, from 60 to 21 seconds per influencer, through workflow optimisation and improved processing logic
- Implemented an error-handling system for analytics, processing records with 99.9% accuracy in identifying and logging processing errors

COMPETITIONS & HACKATHONS

Opensoft'25 - Mindflix | Link **April 2025**
- Pioneered an **AI platform** tackling **5+ well-being concerns** (e.g., stress, burnout) by delivering personalised support recommendations
- Crafted a robust **full-stack solution** using **React/Python**, proactively identifying and addressing **3+ key employee well-being concerns**
- Deployed sophisticated **graph database architecture** using **Neo4j** for relationship mapping and delivering **7+ support recommendations**

Hacktank: Video Proctoring Analysis | Link **March 2025**
- Conceived a **proctoring system** with triple-layer AI/ML combining **visual, audio, and pattern analysis** to **reduce false positives**
- Designed a sophisticated **three-tier analysis architecture** integrating **LLMs**, **rule-based algorithms**, and **machine learning models**

Bhashini Sprint: Voice Controlled Inventory Management | Winner **Jun 2024**
- Devised an **AI-based inventory system** leveraging **LLMs** and **RAG**, processing **natural language commands** across **3+ Indian languages**
- Established a **scalable backend architecture** with **FastAPI** and **Typesense** for search, deployed on **AWS EC2** via **CI/CD Pipeline**
- Received a **Rs. 500K government grant**, demonstrating the platform's potential to revolutionise **inventory management** for **12M+ retailers**

PROJECTS

Personal Hedge Fund - Github Link **Jan 2025 – Apr 2025**
- Integrated **live data streams** from **5+ sources**, automating **ETL pipelines** into a structured database designed for financial analysis
- Constructed **AI-driven agents** to get **financial insights**, enabling informed decisions through **fundamental, technical and sentiment analysis**
- Engineered a robust **backtesting engine**, enabling rapid strategy evaluation alongside **10+ key portfolio management tools** and techniques

Logistics Optimisation for Industrial Expansion - Academic Project **Jan 2024 – Apr 2024**
- Applied **binary linear programming**, **heuristics** and **Clark-Wright algorithm** to optimise warehouse placement across **19 of 150 cities**
- Reduced logistical costs by **$314K**, devising **4 optimal truck routes** using **Cplex solver** integrated with data-driven geographic insights
- Designed a scalable **model accounting for real-world constraints** to enhance company supply chain, projecting **$43.48M in expansion costs**

Self Evaluation of Academic Performance - Research Project **Jun 2023 – Oct 2023**
- Developed a **predictive model** for student GPA using **30+ smartphone data points**, applying comprehensive data cleaning techniques
- Conducted detailed **data analysis** and **20+ feature engineering steps** on collected smartphone data, employing outlier detection methods
- Evaluated **2 different regression models (Lasso, Ridge)** using **K-fold cross-validation**, achieving a low **0.255 Mean Absolute Error**

AWARDS & ACHIEVEMENTS

- Awarded the **Bronze Medal** in the highly competitive **Opensoft General Championship** at **IIT Kharagpur** **Apr 2025**
- Achieved **2nd Place** in the **Hacktank Competition** organised by Optima; awarded a **Rs. 30,000 prize** for the innovative solution **Mar 2025**
- Winner of the government-organised **Bhashini Sprint competition**, securing a **Rs. 500K grant** and **1st place nationwide** **Jul 2024**
- Achieved a ranking within the **Top 5%** among **250,000 candidates** in **JEE Advanced** nationwide **Sep 2022**
- Secured a ranking within the **Top 2%** among over **1.3M candidates** in **JEE Mains** nationwide **Aug 2022**

LEADERSHIP & EXTRACURRICULARS

Executive Head | Quant Club, IIT Kharagpur **May 2023 – Present**
- Spearheaded **research initiatives**, including authoring a comprehensive **white paper** on **algorithmic trading strategies** for club members
- Orchestrated the **WorldQuant BRAIN Consultant Challenge**, collaborating with industry professionals to engage **250+ participants**
- Directed the **"Summer of Quant" initiative**, fostering skill development through **paper trading** and knowledge sharing for **4000+ community**

TECHNICAL SKILLS

Python, C/C++, Scikit-learn, TensorFlow, Deep Learning, NLP, LLMs, RAG, Anomaly Detection, Time Series Analysis, Regression, Classification, Optimization, Operations Research, Algorithmic Trading, Backtesting, Financial Analyst, SQL, PostgreSQL, MongoDB, Neo4j, ETL Pipelines, Data Warehousing, AWS, GCP, Docker, CI/CD, Git, Linux, CLI, CPLEX, Pytorch, Neural Networks